Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/16/2026
PI Name:	Shippen, Dorothy Ph.D.		
Project Title:	Telomere Dynamics and Oxidative Stress in Arabidopsis in the Space Radiation Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology SPACE BIOLOGY--Radiation biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843	**Congressional District:**	10
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	11/21/2022	**End Date:**	11/20/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA KSC
Contact Monitor:	O'Rourke, Aubrie	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	ISS NOTE: End date changed to 11/20/2026 per F. Hernandez/BPS (Ed., 6/3/26).		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Bailey, Susan Ph.D. (Colorado State University) Wyatt, Sarah Ph.D. (Ohio University)		
Grant/Contract No.:	80NSSC23K0302		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose an International Space Station (ISS) Flight Research Investigation using Spectrum hardware, and radiation experiments at NASA Space Radiation Lab (NSRL, Brookhaven) and Colorado State University. This proposal leverages the expertise of three international leaders in the fields of telomere, radiation, and plant biology to explore the relationship between the space radiation environment, oxidative stress, and the dynamics of plant telomeres. Our preliminary data show that, in response to spaceflight, telomerase enzyme activity in Arabidopsis is strongly induced, but telomere length is steady. Telomerase activity was not substantially elevated by simulated microgravity conditions, implying that space radiation is responsible for telomerase induction. Our central hypothesis is that exposure to the space radiation environment results in elevated Reactive Oxygen Species (ROS) which in turn leads to genome oxidation and the high-telomerase activity phenotype observed in the telomere maintenance machinery of space flown plants. We will test this hypothesis with three specific Objectives. Objective 1 will examine the effect of the combined stressors associated with spaceflight on plant telomere dynamics, genome oxidation, and organellar stability in Arabidopsis seedlings grown aboard the ISS as compared to ground controls. We will employ a series of mutant and overexpression lines that influence telomeres and genome oxidation. Objective 2 will explore the impact of space radiation exposure on these parameters using the same Arabidopsis lines by studying the effect of fractionated schemes of simulated space radiation exposure (Galactic Cosmic Ray Simulator at the NSRL) as well as low dose, low dose-rate neutron and low dose gamma-ray exposure (irradiation facilities at Colorado State University). In Objective 3, we synthesize the data gathered from spaceflight and on-ground fractioned radiation exposure experiments to address our central hypothesis. This work is highly impactful as telomeres are an important biological marker of survivability in both plants and humans. Elucidating the relationship between the space radiation environment and telomere dynamics may prove to be a critical component of long-duration missions in the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research addresses how plants endure the impact of the space radiation environment. Our experiments may provide new insight into how different stressors alter the plant genome and the oxidative stress response, and the type of genetic modifications that can be made to enhance survival in harsh environments, including those that currently exist on Earth.
Task Progress:	Telomeres are an important biological marker of survivability. In this project we explore the connection between telomeres, the telomerase enzyme, and oxidative stress for plants exposed to the space radiation environment. Our central hypothesis is that exposure to space radiation increases reactive oxygen species (ROS), which in turn leads to genome oxidation and increased telomerase activity. Radiation-induced telomerase activity then serves a non-canonical role to limit DNA oxidation, thereby protecting the genome. Year 3 saw the completion of all ground-based radiation exposures, the final preflight trial, as well as the completion of the flight itself. Specifically, ground experiments carried out at Colorado State University/CSU (Bailey and Wyatt labs) examined the impact of low dose, low dose rate exposures to neutrons and gamma-rays. Assays conducted in the Shippen lab revealed higher rates of genome oxidation (8-oxoG) after space-relevant radiation exposure, with mutants possessing limited ROS scavenging activity and those without functional telomerase exhibiting higher levels of genome oxidation than plants with these pathways intact. In addition, the final round of ground validation experiments was carried out. With the successful completion of experiment verification testing (EVT), we were approved for launch on SpaceX CRS-32(April 21, 2025). Samples were safely returned to the Shippen lab (Texas A&M AgriLife Research/TAMU) for telomere, telomerase, and genome oxidation measurements. All the proposed assays were completed and the data and residual biological materials were deposited with the Wyatt lab (Ohio University) for final analysis in Year 4. Briefly, we found increased levels of oxidative stress/8-oxoG in flight samples compared to ground controls, with the highest levels in plants lacking telomerase or the ROS scavenger catalase. Notably, plants over-expressing telomerase or the radio-protective Tardigrade protein Dsup showed lower levels of genome oxidation than wild type controls, providing strong evidence to support our central hypothesis.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Fountain LL, Gilliham M, Amitrano C, Arouna N, Barker RJ, Böhmer M, Braun M, Brereton NJB, Brocato RL, Bunchek JM, Canaday ELJ, Caplin N, Castaño P, Chamberlain C, Decourteix M, Del Bianco M, De Micco V, Doherty CJ, Franke MF, Fuentes S, Gilroy S, Harrison L, Hasenstein KH, Hauslage J, Herranz R, Iyer-Pascuzzi A, Izuma DS, Junya K, Kiss JZ, Legué V, Lloyd JPB, Maffei ME, Massa GD, Meyers AD, Perera IY, Poulet L, Roychoudry S, Sena G, Shippen DE, Stoochnoff J, Takahashi H, Wyatt SE, Blancaflor EB. "Expanding frontiers: Harnessing plant biology for space exploration and planetary sustainability." New Phytol. 2025 Nov 26. Online ahead of print. https://doi.org/10.1111/nph.70662 ; PMID: 41295882 , Nov-2025
Articles in Peer-reviewed Journals	Rodriguez A, Barcenilla BB, Hall E, Kundel I, Meyers A, Wyatt S, Shippen D, Kurouski D. "Raman spectroscopy as a tool for assessing plant growth in space and on lunar regolith simulants." npj Microgravity. 2025 May 27;11(1):19. https://doi.org/10.1038/s41526-025-00479-8 ; PMID: 40425596; PMCID: PMC12117163 , May-2025